

A Deep Learning System Outperforms Clinicians in Identifying Optic Nerve Head Abnormalities Heraldng Vision- and Life-threatening Conditions

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Abstract

Background: The examination of the optic nerve head (optic disc) is mandatory in patients with headache, hypertension or any neurological symptoms, yet it is rarely or poorly performed in general clinics. We recently developed a deep learning system (BONSAI-DLS) capable of accurately detecting optic disc abnormalities including papilledema (swelling due to elevated intracranial pressure) on digital fundus photographs with a comparable classification performance to expert neuro-ophthalmologists, but its performance compared to first-line clinicians remains unknown.

Methods: In this international, cross-sectional multicenter study, the DLS, trained on 14341 fundus photographs, was tested on a retrospectively collected convenience sample of 800 photographs (400 normal optic discs, 201 papilledema and 199 other abnormalities) from 454 patients with a robust ground truth diagnosis provided by the referring expert neuro-ophthalmologists. The areas under the receiver-operating-characteristic curves (AUC) were calculated for the BONSAI-DLS. Error rates, accuracy, sensitivity and specificity of the algorithm were compared to those of 30 clinicians with or without ophthalmic training (six internists, six emergency department (ED) physicians, six neurologists, six optometrists and six general ophthalmologists) who graded the same testing set of images.

Findings: The DLS yielded AUCs of 0.97 (CI 95%, 0.96-0.98), 0.96 (CI 95%, 0.94-0.97), and 0.89 (CI 95%, 0.87-0.92) for detection of normal discs, papilledema, and other disc abnormalities, respectively. With an overall error rate of 15.3% (122/800) the BONSAI-DLS outperformed general ophthalmologists, optometrists, neurologists, internists and ED physicians (average 24.4%, 24.8%, 38.2%, 44.8%, 47.9%, respectively). The accuracy of the DLS for the detection of papilledema was significantly higher than that of all clinicians.

Interpretation: The performance of a trained DLS to classify optic discs on photographs was superior to that of first-line clinicians. The BONSAI-DLS could be a valuable diagnostic aid to a broad range of clinicians. Future prospective studies are needed to confirm its applicability in various clinical settings.

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Research in context

Evidence before this study

Optic nerve head (optic disc) abnormalities can herald sight- or life-threatening conditions, requiring urgent detection and treatment. However, the clinicians who manage most often these patients, usually in a context of emergency, are not usually trained in ophthalmology, and are not comfortable with optic disc examination. As an alternative, a limited number of studies using a combination of ocular imaging and machine learning have aimed to differentiate normal optic discs from papilledema (optic disc swelling from intracranial hypertension) on digital ocular photographs. Through the recently established Brain and Optic Nerve Study with Artificial Intelligence (BONSAI) consortium, we were the first to develop a deep learning algorithm (DLS) to detect papilledema and discs with other abnormalities on ocular fundus photographs, with a comparable performance with that of two expert neuro-ophthalmologists. However, expert neuro-ophthalmologists are not the ones who would benefit from such a DLS, as opposed to non-expert clinicians from various specialties, such as neurology, internal medicine, or emergency medicine. Hence, we searched in the literature for a comparison of performance between the DLS and clinicians for the detection of optic disc abnormalities. We searched publications on Pubmed with no date restriction, in English language, with the searched terms “deep learning algorithm” OR “artificial intelligence” AND “humans” OR “clinicians” OR “healthcare providers” AND “optic disc abnormalities” OR “papilledema”. Several studies and few reviews compared deep learning algorithms to experts or non-expert eye-care professionals for the diagnosis of ophthalmic diseases, mostly retinal diseases and glaucomatous neuropathy. They all found that deep learning systems were at least as good as experts and better than non-expert eye-care professionals. However, we found no studies to date that compared the performance of deep learning algorithms to first-line clinicians without training in ophthalmology for the detection of optic disc abnormalities.

In a recent review and meta-analysis, the performance of deep learning algorithms was found equivalent to that of health-care professionals in detecting diseases from medical imaging, although several issues regarding study design, reporting, and clinical value to real world settings were highlighted. Very few studies used the same dataset to compare DLS and human clinicians, reference standards were sometimes poorly defined, diagnostic performance was reported with a wide range of metrics, and only few studies used an external validation dataset to test their algorithm. In another review, the design, reporting standards and risk of bias of various studies comparing the performance of AI and expert clinicians for medical imaging were evaluated; only few studies were prospective and reported transparent data, and the human comparator group was usually too small. In our study, we evaluated the performance of a DLS compared to that of a large number (30) of non-expert clinicians for the classification of images from an independent dataset of normal optic discs, discs with papilledema, and other optic disc abnormalities in neuro-ophthalmic conditions.

Added value of this study

This is the first study to compare the performance of a deep learning system to that of a variety of first-line clinicians without ophthalmic training (six neurologists, six internists, and six emergency department (ED) physicians) and non-expert eye-care professionals (six general ophthalmologists and six optometrists) for the classification of optic disc appearance on the same set of fundus photographs. The reference standard was robust, established by expert neuro-ophthalmologists from the participating centers, and photographs were taken with various cameras in patients with different ethnicities. The overall error rate of the BONSAI-DLS was significantly lower than that of all clinicians, and specifically, the accuracy of the DLS for the detection of papilledema, a sign of sight- or even life-threatening conditions, was significantly higher than that of all clinicians. Such results could be explained by the fact that optic disc examination is a difficult task, maybe not sufficiently taught to medical students during their training. For the detection of normal optic discs, the large majority 26/30 (86.7%) of clinicians were less accurate than the DLS. Clinicians without ophthalmic training (neurologists, internists and ED physicians) critically missed to identify more than a quarter (26.8%) of papilledema images vs. 8.5% for the DLS and falsely diagnosed normal optic discs or optic discs with other abnormalities as papilledema on average 20.6% of the time vs. 4.3% for the DLS. Moreover, the agreement rate amongst clinicians was moderate (0.62) for ophthalmologists and optometrists and low to minimal for the other clinicians (range 0.23-0.45). Our results emphasize the potential utility of a DLS, showing reproducible results for screening of optic disc abnormalities, especially in non-ophthalmic clinical settings, where clinicians without ophthalmic training need to manage those patients and experts are not always readily available.

Implications of all the available evidence

A DLS, trained on a large dataset of mydriatic fundus images originating from multiple centers worldwide, achieved comparable performance to expert neuro-ophthalmologists but significantly outperformed non-expert clinicians for the classification of optic disc appearance on fundus photographs. This is of particular importance for the detection of papilledema, a potential life- and sight-threatening condition often overlooked in non-ophthalmic settings such as neurology, internal medicine or ED departments. This DLS could be a valuable diagnostic aid to a broad range of clinicians in various clinical settings for the screening of optic disc and neurological diseases.

Introduction

Medical imaging has greatly improved the screening and diagnosis of various diseases, including in ophthalmology. Moreover, the recent emergence of artificial intelligence (AI) is expected to provide a valuable additional aid to physicians of various specialties for the interpretation of medical images.¹ Several studies aimed to compare AI and more specifically deep learning (DL), to human clinicians for the interpretation of medical imaging outcomes, in order to evaluate the applicability of that tool in real world settings, and how DL algorithms could affect the clinicians' decision making process.^{2,3} In their review and meta-analysis, Liu and colleagues found that globally, DLs and healthcare providers had a comparable performance.² Yet, several methodology issues were highlighted in most of the studies. These issues included the use of a different datasets for DL and human testing, poorly defined reference standards, or the absence of external datasets for the validation of DL systems.² According to Nagendran *et al*, several biases including lack of data transparency and low reporting standards render the comparison of classification performance by machines and humans unreliable, to date.⁴ In neuro-ophthalmology, as experts are too few and not always readily available,⁵⁻⁷ ocular imaging has recently emerged as an alternative for screening and diagnosis of optic nerve head (optic disc) abnormalities caused by various conditions, including life- or sight-threatening diseases such as cerebral space-occupying lesions and various optic neuropathies, or manifestations of systemic diseases such as malignant hypertension.⁸⁻¹⁰ However, even when fundus photographs are available, the detection and classification of optic disc abnormalities by non-ophthalmology-trained physicians remains suboptimal,^{11,12} and a remote interpretation by neuro-ophthalmologists via telemedicine is usually necessary, but not always effective.^{13,14}

We have recently developed within an international consortium, BONSAI (Brain and Optic Nerve Study with Artificial Intelligence),¹⁵ a deep learning system (BONSAI-DLS) capable of discriminating with high accuracy among normal optic discs, discs with papilledema (optic disc head swelling secondary to raised intracranial pressure), and optic discs displaying other abnormalities (e.g., congenital abnormalities, optic disc drusen, inflammatory, ischemic or atrophic changes) on ocular fundus photographs. Our BONSAI-DLS had a similar performance for the classification of optic disc appearance on ocular fundus photographs as expert neuro-ophthalmologists who did not receive any clinical information along with fundus photographs.¹⁶

To date, no studies compared the performance of a DLS to healthcare providers who are not trained in ophthalmology, but who are supposed to diagnose and manage in first line patients with optic disc abnormalities due to neurological or systemic diseases (e.g. internists, neurologists and emergency department physicians).^{17,18} The aim of this current study was to compare the performance of the BONSAI-DLS to the performance of various clinicians with and without ophthalmic training in classifying optic disc appearance on the same set of fundus photographs previously used to compare the DLS to expert neuro-ophthalmologists,¹⁶ using thorough reporting methods and externally validated data.

Methods

Study design

In this cross-sectional multi-center study, we used a sample of fundus photographs retrospectively collected from the BONSAI Consortium, previously used to compare the BONSAI-DLS to expert neuro-ophthalmologists,¹⁶ to evaluate and compare the classification performance of the BONSAI-DLS and 30 clinicians with and without ophthalmic training. The BONSAI Consortium's objective was to develop, train and test the performance of BONSAI-DLS in classifying optic discs as being normal, displaying features of papilledema or other optic disc abnormalities.¹⁵

The study was approved by the Centralized Institutional Review Board (CIRB) of SingHealth, Singapore and each contributing institution, and was conducted in accordance with the Declaration of Helsinki. Informed consent was exempted given the retrospective nature of the study and the use of de-identified medical information and ocular fundus photographs.

Deep learning system

The BONSAI-DLS classification model has been described in details elsewhere.¹⁵ Briefly, a two-step process was used; first, a segmentation task (using U-Net¹⁹) to automatically detect and crop the optic disc on fundus photographs, and second, a classification task (using DenseNet^{20,21}) to classify the optic disc into one of the three classes: normal, papilledema or other optic disc abnormalities. The BONSAI-DLS was trained and validated on 14,341 photographs from 19 centers in 11 countries, and further externally tested using 1,505 photographs from five other centers.¹⁵

Eligibility criteria, reference standard and retinal imaging in our testing dataset

De-identified standard digital ocular fundus photographs and various demographic and clinical information (reference standard) of patients of different ethnicities with normal optic discs and those with definite neuro-ophthalmic diagnoses affecting the optic discs were provided by neuro-ophthalmology experts from three international reference centers from the BONSAI study (Copenhagen, Denmark; Mayo Clinic Rochester, USA and Seoul, South Korea).¹⁶ The photographs were not used in the training of the BONSAI-DLS and were all taken after pupil dilation, using various digital desktop cameras (Topcon TRC 50DX and TRC NW8, Topcon Medical Systems, Inc, in Copenhagen, Denmark; Topcon TRC 50DX, Topcon Medical Systems, Inc, in Rochester, USA, and Kowa Nonmyd 7 and VX-10, Kowa Company Ltd, Tokyo, Japan in Seoul, South Korea). The reference standard was based on a combination of the optic disc appearance and data derived from medical records, ancillary tests and follow-up visits. Inclusion criteria for normal optic discs included normal vision, normal clinical examination and normal ancillary tests (if applicable), in the absence of other eye disease except for mild cataract. Papilledema was defined as optic disc swelling associated with raised intracranial pressure confirmed by brain imaging (brain tumor or cerebral venous sinus thrombosis) and/or elevated cerebrospinal fluid opening pressure. Patients with papilledema were diagnosed with idiopathic intracranial hypertension when the modified Dandy

criteria were met.²² Optic discs with other abnormalities were defined as harboring optic neuropathies (inflammatory or ischemic), optic atrophy, optic disc drusen or other congenital abnormalities, with the diagnosis being made according to standard clinical and ancillary testing criteria⁹ (e.g. optical coherence tomography, ultrasound, fundus autofluorescence, magnetic resonance imaging, etc.). All fundus photographs provided by the three participating centers were reviewed by two neuro-ophthalmologists (DM and CV), and after exclusion of images of poor quality or with incomplete data or mislabeling, a convenience sample of 800 out of 1347 fundus photographs (composed of half normal and half abnormal optic discs) from 454 patients (346 patients with both eyes imaged and 108 singletons with one eye imaged),¹⁶ was utilized (**table 1**).

Sample size calculation

For $\alpha = 0.05$; a prevalence of papilledema of 25.1% (201/800 images) in our retrospectively collected testing dataset used in a previous study;¹⁶ a predicted DLS sensitivity and specificity of 96.4% and 84.5%, respectively,¹⁵ and a predicted minimum (i.e., attenuated by 40% compared to the DLS) clinician sensitivity and specificity of 56.4% and 44.5%, respectively, with a maximum marginal error of 7%, the sample size for adequate testing of the sensitivity and specificity of the DLS and clinicians for the detection of papilledema was 767 eyes including 575 control/other eyes and 193 eyes with papilledema.²³

Clinicians' selection and testing procedure

Thirty clinicians of different levels of expertise from five different medical specialties participated in this study. They were asked to provide a description of optic disc appearances, i.e. papilledema, other optic neuropathies or normal for each fundus photograph. Eighteen clinicians were physicians without ophthalmic training, naïve to fundus photographs reading, with some experience in optic disc examination, including six internists (one resident, one senior resident and four senior consultants), six ED physicians (one senior resident, one first year junior consultant, and four senior consultants) and six neurologists (one resident doing general neurology, one senior resident and four senior consultants), all practicing in the Singapore General Hospital, Singapore. Twelve clinicians had ophthalmic training: six optometrists (all practicing for more than five years in an academic reference center), and six general ophthalmologists (two residents and four general ophthalmology consultants), all practicing in the Singapore National Eye Center, Singapore. None of them had fellowship level training or specific expertise in neuro-ophthalmology.

The 800 fundus photographs were randomly and independently presented to the different clinicians on the same individual computer screen (LG-34WK650, 100% brightness, 80% contrast), as previously done for expert neuro-ophthalmologists.¹⁶ A semi-automated software²⁴ developed by LM and RPN was used to randomly display fundus photographs and combine the grading provided by the clinicians in an excel sheet. Prior to being tested, all clinicians received a short training session on how to use the software on 50 fundus photographs illustrating all categories of optic disc findings. Subsequently, without any clinical information, each clinician was independently given the fundus photographs and

classified the optic discs as normal, papilledema, or other optic disc abnormalities. The time spent to complete the classification of 800 fundus photographs was recorded for each clinician.

Statistical analyses

The performance characteristics of the BONSAI-DLS were determined through the calculation of the overall multiclass error rate percentage (i.e. the rate of incorrect overall classification compared to the reference standard), the area under the receiver operating characteristic curve (AUC) as well as sensitivity, specificity, and accuracy, using a one-vs-rest strategy (normal versus the rest, papilledema versus the rest, and other abnormalities versus the rest). Thresholds were pre-set to achieve high sensitivity (at least 90%) on the BONSAI study primary validation datasets.¹⁵ The performance characteristics of clinicians were determined through the calculation of the overall multiclass error rate percentage, as well as sensitivity, specificity, and accuracy using a one-vs-rest strategy.

A pairwise McNemar test with Bonferroni correction was used to compare the sensitivities, specificities and accuracies between the BONSAI-DLS and each clinician.²⁵ The percent error rate was compared between ophthalmology-trained and non-ophthalmology-trained healthcare providers using a Mann Whitney U-Test. The percent error rate was also compared between specialty groups using a Kruskal Wallis analysis of variance on ranks followed by a pairwise comparison using a Tukey posthoc test.

Analyses were performed using a modified version of the package “DTComPair” in R v.3.6.3: A Language and Environment for Statistical Computing (R Core Team, Vienna, Austria). The multi-class classification inter-grader agreements (within each group of clinicians) were computed using Cohen’s kappa agreement scores. Confidence intervals (95% CI) were calculated for the AUC, sensitivity, specificity, accuracy and kappa scores. Unless indicated otherwise, data are represented in the text, tables and figures as average (95% CI).

Results

Demographics and fundus photographs characteristics

The study population included 454 patients, mean age was 44.5 (± 20.1) years, and 273/454 (60.1%) were female (**table 1**). Half of the fundus photographs were normal (400/800, 50.0%) and the other half was divided into papilledema (201 images, 25.1%), and other optic disc abnormalities (199 images, 24.9%) (**table 1**). Details regarding demographics and number of patients and fundus photographs per optic disc category and center are described in **table 1**.

Error rates of BONSAI-DLS and clinicians for overall classification

As previously reported¹⁶ the overall error rate of the BONSAI-DLS for the classification of optic disc appearance on this sample of 800 photographs was 15.3% (122/800), which was a better performance than any of the clinicians in this study, including those with ophthalmic training (**figure 1A and 1B**).

The overall error rates of the clinicians were 24.4% for ophthalmologists, 24.8% for optometrists, 38.2% for neurologists, 44.8% for internists, and 47.9% for ED physicians, all statistically higher

than the error rate of the BONSAI-DLS (**figure 1B**, $P < 0.001$). Ophthalmologists and optometrists displayed a lower error rate than physicians without ophthalmic training (**figure 1A**, 24.6% vs. 43.6%, $P < 0.001$). The performance of ophthalmologists and optometrists was comparable (**figure 1B**, $P = 1$).

The BONSAI-DLS was on average 168 times faster in the classification of 800 fundus photographs compared to clinicians (average times: BONSAI-DLS = 25 seconds; clinicians = 70 minutes, STD ± 22.3 min, range 40-139 min).

Comparison of performance (AUC, accuracy, sensitivity and specificity) between BONSAI-DLS and clinicians

The BONSAI-DLS had an AUC of 0.96 (CI 95% 0.94-0.97), 0.97 (CI 95% 0.96-0.98) and 0.89 (CI 95% 0.87-0.92) for the detection of papilledema, normal optic discs and other optic disc abnormalities, respectively¹⁶. The overall performances of clinicians for the detection of papilledema, normal optic discs and other optic disc abnormalities were all below the ROCs of the DLS, indicating a lower classification performance overall (**figure 2**).

Of the 30 clinicians, 30 (100%), 26 (86.7%) and 28 (93.3%) had a significantly lower accuracies than the DLS for the grading of papilledema, normal optic discs and other optic disc abnormalities, respectively (**figure 3, table 2, supplementary Tables 1 and 2**). Three ophthalmologists and one optometrist were as accurate as the DLS for the detection of normal optic discs, and one ophthalmologist and one optometrist were as accurate as the DLS for the detection of other optic disc abnormalities.

For the detection of papilledema, 11/30 (36.7%) clinicians were significantly less sensitive than the DLS, while 28/30 (93.3%) were less specific. The results of sensitivity and specificity are oppositely interdependent. Hence, some internists, neurologists and ED physicians displayed relative high sensitivities, at the detriment of a high false positive rate (i.e. low specificity, **figure 2, supplementary Figure 1, table 2**).

For the detection of normal optic discs and other optic disc abnormalities, 24/30 (80%) and 30/30 (100%) clinicians were less sensitive than the DLS, and 8/30 (26.7%) and 16/30 (53.3%) were less specific, respectively (**figure 2, supplementary Figure 1, supplementary Tables 1 and 2**).

Inter-agreement within clinician groups

Inter-grader agreements for ophthalmologists, optometrists, neurologists, internists and ED physicians followed a decreasing trend, from moderate (0.62) for ophthalmologists and optometrists to minimal agreement (0.25) for the ED physicians (**supplementary Figure 2**).²⁶

Discussion

The main finding of this study is that a DLS outperformed non-expert clinicians from various medical specialties, including non-ophthalmology-trained physicians and eye-care professionals, in the detection of optic nerve head abnormalities on ocular fundus photographs. The DLS, developed on

data originated from a large, international, multicenter consortium, was more accurate than any individual clinician who performed the same task for the detection of papilledema, a sign of often life or sight-threatening conditions. Additionally, the DLS performed better than most (86.7% and 93.3%, respectively) clinicians for the detection of normal optic discs and other optic disc abnormalities. Not surprisingly, overall accuracies and intergrader agreements were higher for eye-care professionals than for physicians without ophthalmic training.

Deep learning algorithms are increasingly developed to detect systemic biomarkers or retinal diseases associated with general conditions using fundus photographs acquired with digital cameras.^{27–30}

However, only a few studies used deep learning techniques to detect optic disc abnormalities related to neuro-ophthalmic conditions.¹⁷ Optic disc abnormalities can affect a substantial number of patients with severe hypertension, headaches or neurologic deficits who present to non-ophthalmic providers.^{31,32} Identification of one of the three categories of optic disc findings described in our study (normal, papilledema and other optic disc abnormalities) should generate different acute management and patient disposition, as correct recognition of normal optic discs should be reassuring and defer referral to specialists, while findings of papilledema should prompt immediate neuroimaging with or without lumbar puncture. To our knowledge, there is no other study comparing the performance of a deep learning algorithm to that of non-ophthalmology trained physicians, such as Neurologists, Internists or Emergency physicians, for the detection of optic disc abnormalities. In this study, the BONSAI-DLS accurately detected papilledema on fundus photographs in 91.5% of cases. In contrast, non-ophthalmology-trained physicians (i.e. internists, ED physicians and neurologists) evaluating fundus photographs without any clinical information, missed 26.8% of papilledema on average vs. 8.5% for the DLS, 16% on average for non-expert eye-care professionals (i.e. general ophthalmologists and optometrists), and 10.7% for expert neuro-ophthalmologists.¹⁶ Moreover, non-ophthalmologists on average falsely diagnosed normal optic discs or optic discs with other abnormalities as papilledema 20.6% of the time vs. 4.3% for the DLS, 11.1% on average for eye-care professionals and 5.9% for expert neuro-ophthalmologists.¹⁶ High error rates and false positives for detection of papilledema could potentially lead to diagnostic error or delays, or to unnecessary and costly tests or referrals.^{33,34}

Comparing DLS to clinicians is an important step before a potential implementation of automated screening or diagnosis of diseases in real world clinical practice. In ophthalmology, some studies evaluated the detection of glaucomatous optic neuropathies on ocular fundus photographs, and showed that DLS performance was at least as good as that of expert human evaluators and better than that of non-experts.^{18,35–37} Likewise, classification of diabetic retinopathy, retinopathy of prematurity or age-related macular degeneration was more accurately achieved by various DLSs, with performance similar or better than that of human evaluators.^{2,38–44} However, two recent reviews highlighted several concerns regarding study design, reporting standards and risk of bias of many studies that compared the performance of DLS to human clinicians for interpretation of medical

imaging.^{2,4} Liu *et al* included 25 studies in a meta-analysis, among them six in ophthalmology, and found an equivalent performance between DLS and health-care professionals for the classification of diseases using medical imaging.² Nevertheless, only six studies out of twenty-five were prospective, the reference standard was usually not well-defined, and only 14 used the same sample of images to test the DLS and human clinicians.² Nagendran *et al* found that 72% of non-randomized studies comparing DLS to human clinicians were at high risk of bias, partly due to a small number of human comparators (median of five clinicians), and model performance measures that were not appropriately evaluated.⁴ In this study, we used a DLS previously trained on 14341 fundus photographs in a large multicentre study, and externally validated. The reference standard was robust, established by expert neuro-ophthalmologists from the participating centers, and the human comparator group was composed of 30 clinicians, of various specialties, and grouped into six clinicians by specialty. The same dataset of 800 fundus photographs was utilized to compare the BONSAI-DLS, two expert neuro-ophthalmologists in a previous study,¹⁶ and the 30 clinicians included in the current study for the classification of optic disc appearance on fundus photographs. The BONSAI-DLS outperformed all the clinicians, including eye-care professionals (optometrists and general ophthalmologists) who did not have a specific expertise in optic disc diseases. Yet, optometrists are increasingly involved in the screening and management of ocular diseases and thus, are often the first-line providers for the diagnosis of optic disc diseases.⁴⁵

In our study, inter-grader agreements were moderate for ophthalmologists and for optometrists (0.62 for both), and minimal to weak for non-ophthalmology-trained physicians (0.29, 0.25 and 0.45 for internists, ED physicians and neurologists, respectively). A lack of agreement among experts for interpretation of ocular imaging has been previously described.^{40,42} In one study, eight experts who were evaluating optical coherence tomography (OCT) findings to determine the need for urgent referral of various retinal conditions reached only a 65% agreement rate.⁴⁰ Another study showed a wide variability of inter-agreement scores (ranging from 0.59 to 0.92) for the diagnosis of plus disease retinopathy of prematurity from retinal fundus photographs among nine experts.⁴² This high variability of interpretation of fundus images amongst groups of physicians of equivalent level of expertise emphasizes the need for a reproducible deep learning algorithm for screening purposes. In a recent study, the use of a machine learning-based decision support system was associated with an increased interobserver agreement and increased performance for less experienced participants.³ This study has inherent limitations. First, the clinicians were not given any clinical information along with the fundus photographs, which is not representative of real-life clinical settings and could have affected their performance.^{2,46} Indeed, the fundus photographs were provided to us along with the ground truth diagnosis established by the participant center's expert, yet without any detail regarding the clinical presentation of the patients. Second, for the purpose of comparing the DLS to humans' performance in classifying optic disc findings, we selected a retrospective convenience sample composed of comparable distribution of each category of optic disc abnormalities in the different

datasets (including approximately 50% normal discs, 25% discs with papilledema and 25% other optic disc abnormalities). This is not the typical condition prevalence seen in clinical practice. However, the fundus photographs for each category were randomly collected, and the clinicians were not aware of the diagnosis distribution before being tested, likely reducing the risk of bias in their interpretation of the images. To mitigate such potential bias, future prospective studies should include consecutive patients, selected from real-life conditions, with additional clinical information along with ocular imaging. Third, all the clinicians who participated in this study were practicing in the same hospital group, the Singapore General Hospital. Although most of them were trained abroad, this could represent a bias to the clinicians' performance.

In conclusion, an AI-based DLS outperformed 30 clinicians for the classification of optic disc appearance on ocular fundus photographs, especially for the detection of papilledema, a potentially life- and sight-threatening condition not infrequently encountered in non-ophthalmic settings. The BONSAI-DLS could be a valuable diagnostic aid in medical and referral decision-making when patients present to internal medicine, neurology or the ED with visual loss, headaches, neurologic symptoms, or severe hypertension. Such a diagnostic aid could be particularly valuable in low-income countries or regions that lack immediate accessibility to ophthalmologists. It also has the potential to influence management and improve value based driven care and reduce unnecessary imaging and treatment in developed healthcare systems. Future prospective studies in real-life settings are needed to confirm the applicability of this promising tool.

Contributors

CV, LM, DT, NJN, VB, DM and RPN contributed to the conception and design of the study. CV, XX, JMH, SH, JJC, JLL, LM, KT, YL, NJN, VB, DM and RPN had a major role in the acquisition of the data. CV, SN, XX, YL, NJN, VB, DM and RPN contributed to the analysis and/or interpretation of the data. CV, DM and RPN contributed to writing the manuscript, and CV, SN, XX, YL, NJN, VB, TYW, DM and RPN contributed to the revision of the manuscript for intellectual content. All authors approved the manuscript. CV, DM and RPN guarantee the integrity of the work. CV and RPN contributed equally to this work.

Declaration of interests

None of the authors have conflict of interests to declare.

Data sharing statement

Deidentified data will be made available by reasonable request to the corresponding author (email address: dan.milea@singhealth.com.sg).

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Figures legend

Figure 1 Parts A-B: Overall error rates of the deep learning system (dashed grey line) and clinicians for the classification of optic disc appearance.

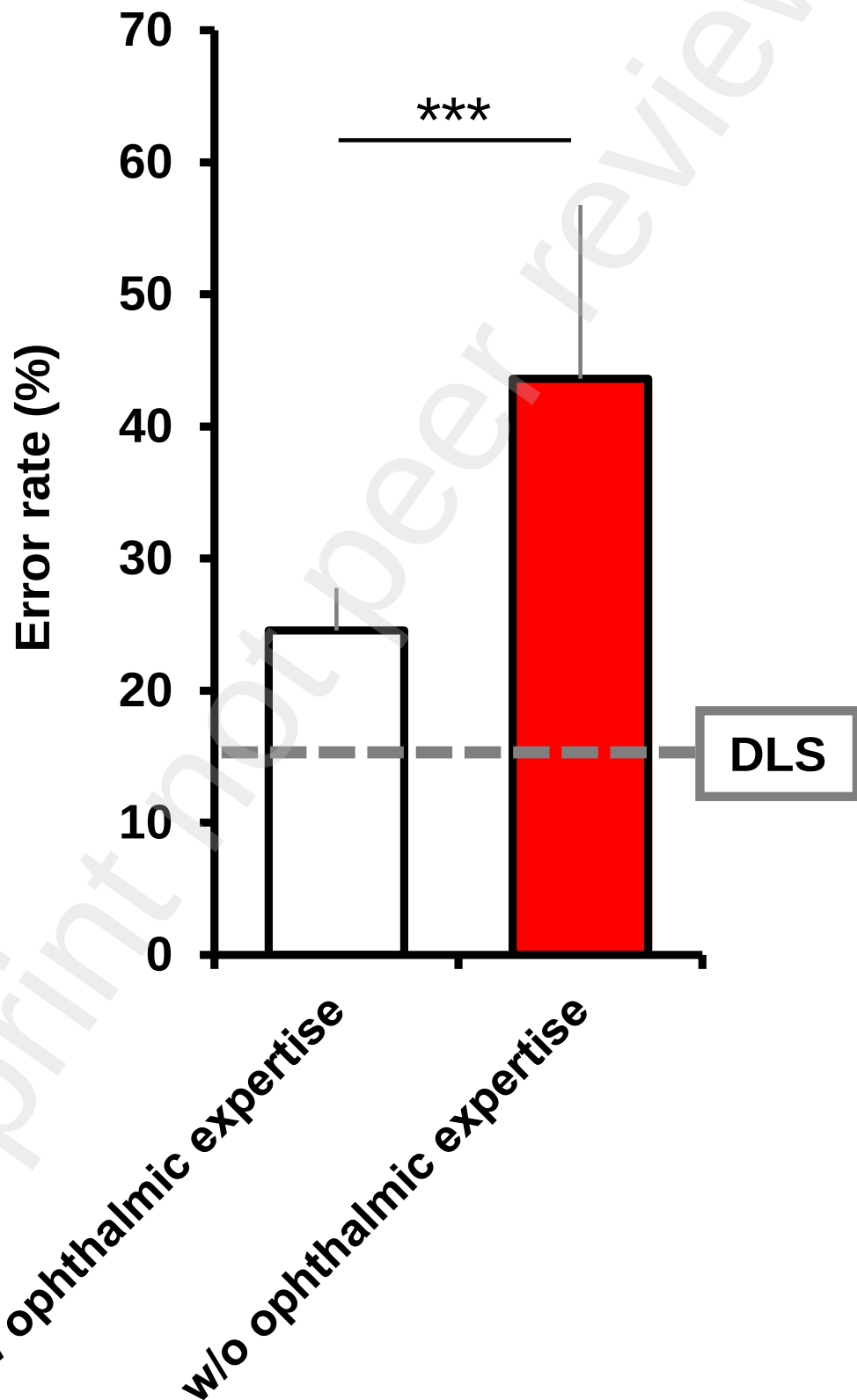
A- Clinicians are grouped according to their training in ophthalmology. B- Detailed results for each group of specialists, from the highest (left) to the lowest (right) performance. Data are represented in percentage, as average $\pm$ standard deviation. The error rate of the deep learning system was significantly lower than that of all the clinicians (A and B). Physicians without ophthalmic expertise had significantly higher error rates than ophthalmologists and optometrists (A- Mann Whitney U-Test ($P < 0.001$) and B- Kruskal-Wallis One Way ANOVA on Ranks ($H = 20.67$, $P < 0.001$)). Tukey test was used for post hoc analyses.

Figure 2 Parts A-C: Receiver operating characteristic curve (ROC) corresponding to the performance of the deep learning system (BONSAI-DLS) and clinicians (n=30) for the classification of optic discs on fundus photographs.

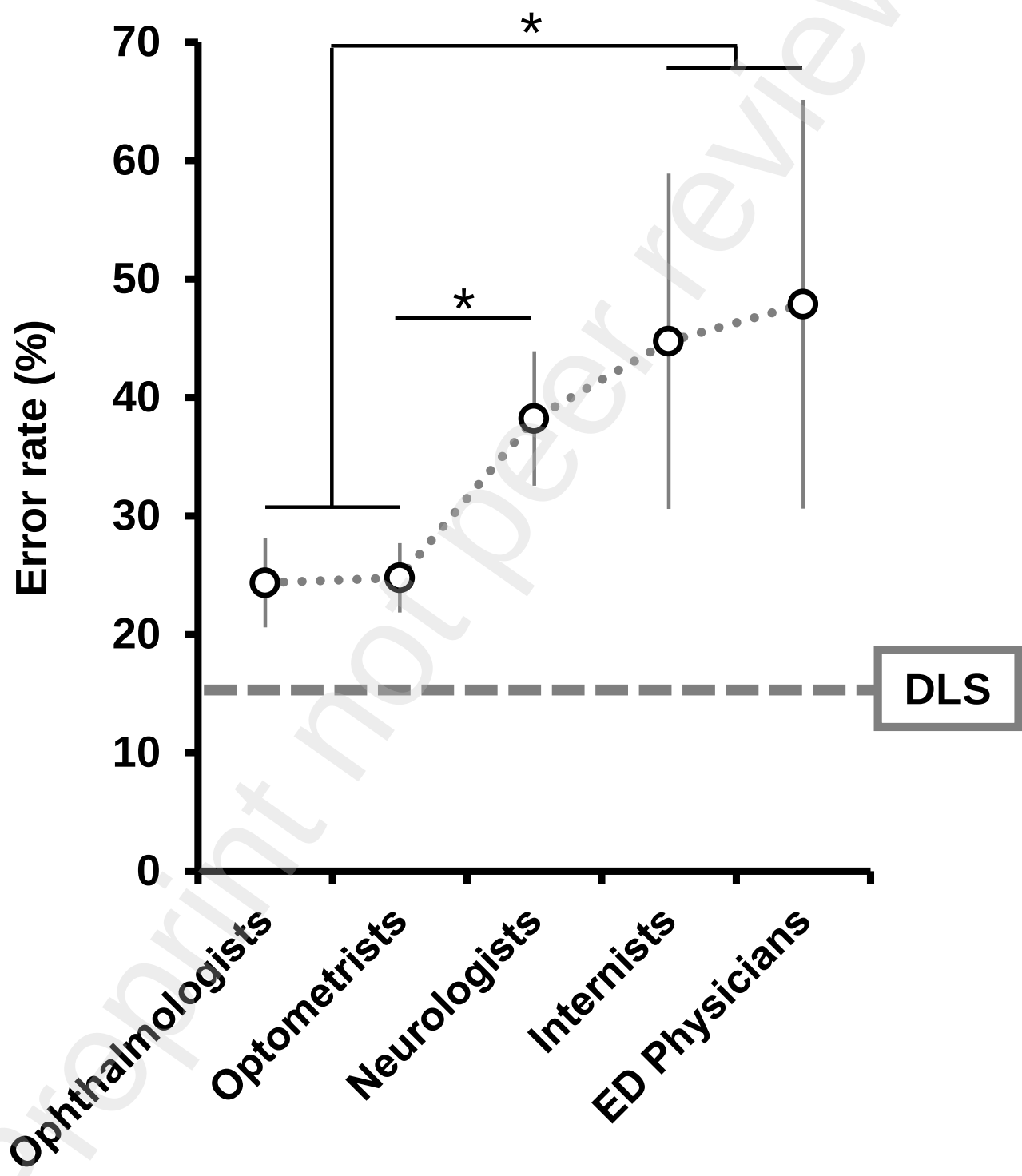
A- Normal optic discs. B- Discs with papilledema C- optic discs with other abnormalities. The results for clinicians are presented as average and standard deviation for each group of specialists. The DLS had an AUC of 0.97 (95%CI 0.96-0.98) for normal optic discs, 0.96 (95% CI 0.94-0.97) for papilledema and 0.89 (95% CI 0.87-0.92) for other optic disc abnormalities. For more details on the performance characteristics of the DLS and clinicians, see table 2 and appendix tables 1 and 2.

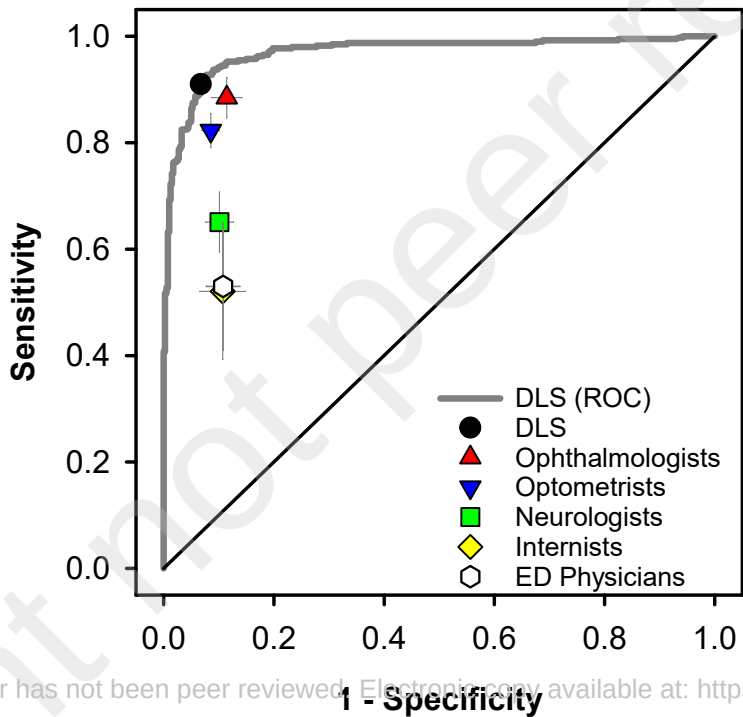
Figure 3: Accuracy of the DLS and clinicians for the detection of normal discs, discs with papilledema or other abnormalities. The grey area represents the 95% confidence interval of accuracy of the DLS, and the dashed gray line the average. Blue boxplots represent the mean and standard deviation of accuracies in each group of clinicians. Each evaluator is represented by a red circle.

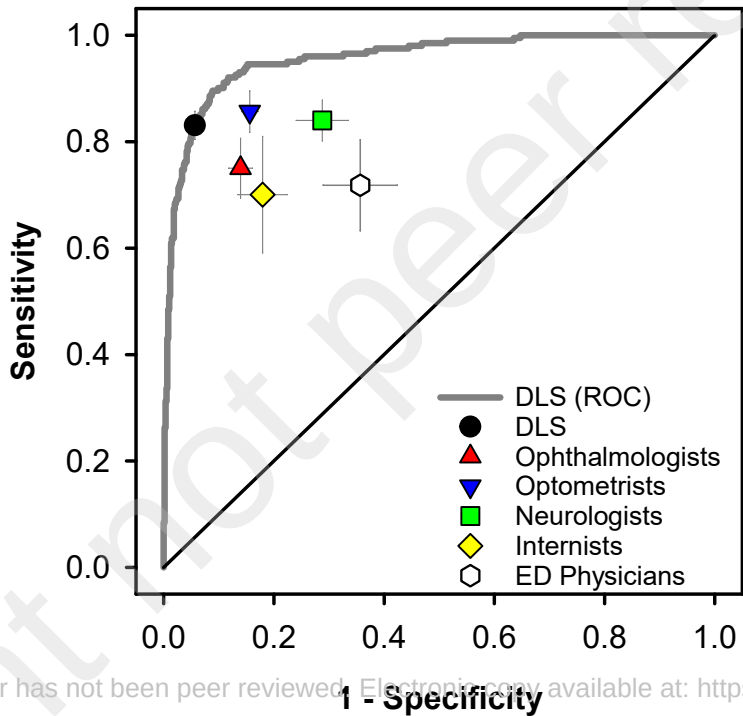
A.



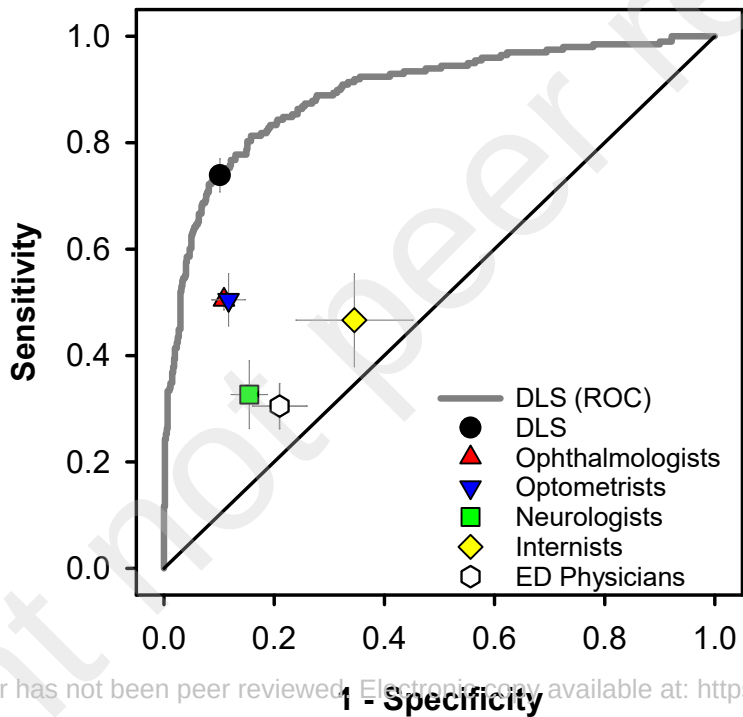
B.



A.

B.

C.



Accuracy

Normals

Papilledema

Others

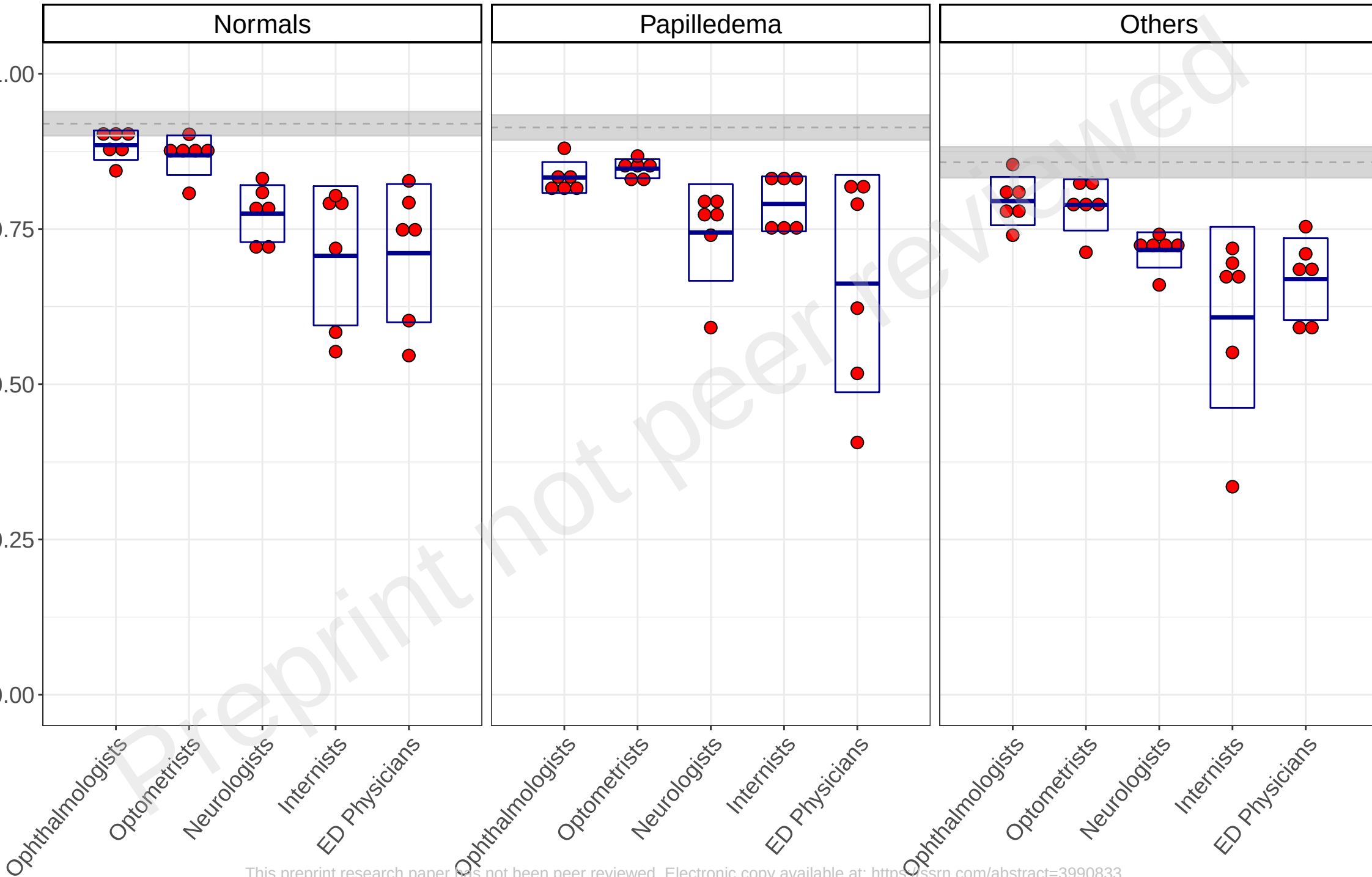


Table 1: Demographics and characteristics of patients and imaged eyes per center

	Copenhagen	Rochester	Seoul	Overall
Patients characteristics				
Mean age $\pm$ STD (years)	40.1 $\pm$ 16.9	50.9 $\pm$ 18.6	43.0 $\pm$ 21.2	44.5 $\pm$ 20.1
Female sex (%)	73.7	62.6	53.0	60.1
Race (%)				
Asian	0	0	99.6	48.7
Caucasian	100	98.4	0.4	50.9
Other	0	1.6	0	0.4
Number of patients (imaged eyes)				
Normal optic discs	46 (86)	58 (92)	131 (222)	235 (400)
Papilledema	23 (45)	26 (50)	53 (106)	102 (201)
Other optic disc abnormalities	30 (57)	39 (66)	48 (76)	117 (199)
Total patients (eyes)	99 (188)	123 (208)	232 (404)	454 (800)

STD: standard deviation

Table 2: Performance of the deep learning system and clinicians for the classification of discs with papilledema from normal discs and discs with other abnormalities

Classification of discs with papilledema			
	Sensitivity (%) ^a	Specificity (%)	Accuracy (%)
DLS	83.1 (77.9-88.3)	94.3 (92.5-96.2)	91.5 (89.4-93.3)
Ophthalmologist 1	76.6 (70.8-82.5)	82.5 (79.4-85.5) *** ^b	81.0 (78.1-83.7) ***
Ophthalmologist 2	66.2 (59.6-72.7) ***	87.5 (84.8-90.1) ***	82.1 (79.3-84.7) ***
Ophthalmologist 3	52.2 (45.3-59.1) ***	94.0 (92.1-95.9)	83.5 (80.7-86.0) ***
Ophthalmologist 4	90.5 (86.5-94.6) *	80.8 (77.6-84.0) ***	83.3 (80.5-85.8) ***
Ophthalmologist 5	82.6 (77.3-87.8)	81.6 (78.5-84.7) ***	81.9 (79.0-84.5) ***
Ophthalmologist 6	82.1 (76.8-87.4)	90.0 (87.6-92.4) ***	88.0 (85.5-90.2) **
Optometrist 1	85.6 (80.7-90.4)	85.6 (82.8-88.5) ***	85.6 (83.0-88.0) ***
Optometrist 2	88.6 (84.2-93.0)	86.1 (83.4-88.9) ***	86.8 (84.2-89.0) **
Optometrist 3	68.2 (61.7-74.6) ***	90.3 (87.9-92.7) **	84.8 (82.1-87.2) ***
Optometrist 4	89.1 (84.7-93.4)	81.8 (78.7-84.9) ***	83.6 (80.9-86.1) ***
Optometrist 5	85.6 (80.7-90.4)	81.3 (78.2-84.4) ***	82.4 (79.6-85.0) ***
Optometrist 6	97.0 (94.7-99.4) ***	81.1 (78.0-84.3) ***	85.1 (82.5-87.5) ***
Neurologist 1	66.7 (60.1-73.2) **	80.0 (76.8-83.2) ***	76.6 (73.5-79.5) ***
Neurologist 2	87.1 (82.4-91.7)	76.5 (73.1-79.9) ***	79.1 (76.1-91.9) ***
Neurologist 3	84.6 (79.6-89.6)	70.5 (66.8-74.1) ***	74.0 (70.8-77.0) ***
Neurologist 4	93.5 (90.1-96.9) ***	72.8 (69.2-76.4) ***	78.0 (75.0-80.8) ***
Neurologist 5	82.1 (76.8-87.4)	79.0 (75.7-82.2) ***	79.8 (76.8-82.5) ***
Neurologist 6	90.0 (85.9-94.2) *	48.7 (44.7-52.8) ***	59.1 (55.6-62.6) ***
Internist 1	94.5 (91.4-97.7) **	68.1 (64.4-71.8) ***	74.8 (71.6-77.7) ***
Internist 2	70.7 (64.4-76.9) *	87.0 (84.3-89.7) ***	82.9 (80.1-85.4) ***
Internist 3	19.4 (13.9-24.9) ***	95.0 (93.2-96.7)	76.0 (72.9-78.9) ***
Internist 4	72.6 (66.5-78.8) **	87.1 (84.5-89.8) ***	83.5 (80.7-86.0) ***
Internist 5	90.5 (86.5-94.6) *	68.9 (65.2-72.7) ***	74.4 (80.7-86.0) ***
Internist 6	72.6 (66.5-78.8) **	86.1 (83.4-88.9) ***	82.8 (79.9-85.3) ***
ED physician 1	71.6 (65.4-77.9) *	59.1 (55.2-63.0) ***	62.3 (58.8-65.6) ***
ED physician 2	51.2 (44.3-58.2) ***	51.9 (47.9-56.0) ***	51.8 (48.2-55.3) ***
ED physician 3	88.6 (84.2-93.0)	75.8 (72.4-79.2) ***	79.0 (76.0-81.8) ***
ED physician 4	87.1 (82.4-91.8)	79.5 (76.2-82.7) ***	81.4 (78.5-84.0) ***
ED physician 5	41.8 (35.0-48.6) ***	40.2 (36.3-44.2) ***	40.6 (37.2-44.1) ***
ED physician 6	90.5 (86.5-94.6) *	79.5 (76.2-82.7) ***	82.3 (79.4-84.8) ***

DLS: deep learning system, ED: emergency department

^a Data are represented as mean (95% CI). 95% CI were calculated using the asymptotic method.

^b Statistical significance DLS vs clinicians *: P < 0.05; **: P < 0.01; ***: P < 0.001.